

Hawkeye EQMS · Deviation / Non-Conformance · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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1. Module overview

Module	Deviation / Non-Conformance
Pharma purpose	Capture, investigate, disposition and close any planned or unplanned deviation from a specification, SOP, batch record, or quality-system requirement. Drive a CAPA when the deviation is systemic.
Compliance basis	ISO 9001:2015 §10.2 (non-conformity & corrective action) · 21 CFR 211.192 (production record review · investigations) · ICH Q10 §3.2.3 (deviation management) · EU GMP Chapter 1.4 (quality system).
Hawkeye routes	<code>/nonconformance</code> (frontend) · <code>/api/deviations</code> (backend)
Hawkeye model file	<code>backend/src/models/DeviationModel.js</code>
Tenant module flag	<code>modules.EVENT_MANAGEMENT</code>

A 'deviation' is any departure from the approved batch record, validated procedure, or specification. Pharma QA teams treat the deviation as the entry-point of the quality investigation lifecycle: it determines whether the affected batch can be released, whether a CAPA is required, and (if patient-safety / regulatory exposure) whether a Field Alert Report or recall must be filed. Hawkeye stores deviations in the `deviations` collection, exposes them to users with the `EVENT_MANAGEMENT` module flag enabled, and offers AI assistance at the investigation step (Wave 1 · 5-Why scaffolder + CAPA RCA drafter) and at the disposition step (Wave 3 · predictive CAPA outcome).

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Single intake form for any operator to log a deviation 21 CFR 211.192	+ Report Deviation dialog on /nonconformance with title, description, type (PLANNED / UNPLANNED), classification (CRITICAL / MAJOR / MINOR), category (10 enums), area, processStep, productName, immediateActions	MET	
Auto-numbering of deviation records (DEV-YYYY-NNNN) GMP traceability	deviationNumber auto-generated server-side (DeviationModel.js:54). Format DEV-YYYY-NNNN.	MET	
Multi-step lifecycle with role gates (operator → QA → Head of QA → VP) ICH Q10 §3.2.3	9-state machine (REPORTED → UNDER_ASSESSMENT → UNDER_INVESTIGATION → PENDING_DISPOSITION → PENDING_CAPA_DECISION → CAPA_REQUIRED → PENDING_CLOSURE → CLOSED, plus CANCELLED). Transitions gated by EDITOR_ROLES.	MET	
Impact assessment captured before investigation (product / patient / batch / regulatory) 21 CFR 211.192	POST /assess endpoint requires productQualityImpact, patientSafetyImpact, batchDisposition, regulatoryImpact (deviationRoutes.js:104-124)	MET	
Root-cause analysis with method tracking (5-Why / Fishbone / FTA / FMEA / Pareto) ICH Q9	POST /investigate accepts method enum + summary + rootCause + rootCauseCategory. AI 5-Why scaffolder available in the View+AI drawer.	MET	
Batch disposition decision (RELEASE / REJECT / REWORK / REPROCESS / QUARANTINE) 21 CFR 211.165	POST /dispose with decision + justification. Five-value enum.	MET	
Decision whether CAPA is required, with auto-creation of CAPA record ISO 9001 §10.2.1	POST /capa-decision with capaRequired flag. autoCreateCapa=true triggers crossModuleService.createCapaFromDeviation() — produces a real capa-v2 record, not just a flag.	MET	
Closure with electronic signature meeting 21 CFR Part 11 21 CFR Part 11 §11.50	POST /close captures closureNotes + closedBy + closedAt. Generic /api/electronic-signatures/sign endpoint exists and can be wired to closure but is NOT enforced today.	PARTIAL	Wire e-sig requirement to closure (1-line check)

Pharma expectation	Hawkeye implementation	Status	Note
Linked records: CAPAs, complaints, change controls, audits ICH Q10 §3.2.5	linkedCAPAIds, linkedComplaintIds, linkedChangeControlIds, linkedAuditIds, linkedDeviationIds arrays on the model. Cross-module endpoint POST /api/quality/capa-from-deviation creates + links.	MET	
Field Alert Report / MDR reportability flag 21 CFR 314.81 / 21 CFR 803	regulatoryImpact captured as free-text on the assessment but no MDR / FAR boolean / deadline timer.	GAP	Add isFieldAlertReportable + farDueDate + auto-reminder
Trend / signal detection across deviations (e.g. 3 OOS in 30 days on same equipment) ICH Q10 §3.2.4 (KPI monitoring)	Wave 3 SignalAlertsList — Z-score cluster detector. Live alert today: equipment:NVX-PRESS-001 · size 3 · z=3.4.	MET	
AI assistance to accelerate RCA + draft CAPA + predict outcome FDA AI guidance Jan 2025	View + AI drawer offers Scaffold 5-Why (Wave 1) + Draft CAPA RCA (Wave 1) + Predictive CAPA badge (Wave 3) + Create CAPA from RCA cross-module button.	MET	
Audit trail of every state change + every AI accept/reject (ALCOA+) 21 CFR Part 11 §11.10(e)	auditTrailModel logs deviation transitions. AI outputs logged to AiActionMetric + DataIntegrityLog with prompt-version + retrieval-set hashes.	MET	
Per-tenant module enable / disable Multi-tenant SaaS	tenantModuleConfig with EVENT_MANAGEMENT boolean. Frontend hides nav + 403s when disabled.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Kenji Tanaka QA Specialist (admin)	<code>qa.specialist@novex-pharma.demo</code>	Reports deviations from the floor, runs the initial impact assessment, drives investigation + RCA, accepts AI-drafted RCA, creates CAPA from drawer.	REPORTED UNDER_ASSESSMENT UNDER_INVESTIGATION
James Thompson Head of QA (admin)	<code>qa.head@novex-pharma.demo</code>	Reviews investigation conclusions, makes batch disposition decision, decides whether systemic CAPA is required, signs closure.	PENDING_DISPOSITION PENDING_CAPA_DECISION CAPA_REQUIRED PENDING_CLOSURE CLOSED

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Elena Vasquez VP Quality (tenant_admin)	vp.quality@novex-pharma.demo	Final approver on CRITICAL deviations; reviews closure on the management-review cycle. Signs e-sig where enforced.	PENDING_CLOSURE CLOSED
Marcus Brown Regulatory Affairs (admin)	regulatory@novex-pharma.demo	Determines MDR / Field Alert reportability when deviation has patient-safety or regulatory exposure.	UNDER_ASSESSMENT UNDER_INVESTIGATION
Maria Santos Lead Auditor (auditor)	audit.lead@novex-pharma.demo	Reviews deviation closure during periodic internal audit. Triage CAPAs spawned from deviations.	CLOSED (audit-time review)

4. Feature catalogue (click-by-click)

4.1 View deviation register

What it does	Land on /nonconformance to see every deviation in the tenant with classification + status chips and inline workflow buttons.
Menu / route	Top nav · 'Deviations' tile · /nonconformance
Available to roles	admin buyer auditor tenant_admin
Backend API	GET /api/deviations?page=1&limit=20

CLICK-BY-CLICK

- NAVIGATE** Click 'Deviations' in the top nav (or open /nonconformance directly)
→ Page loads with header 'Deviation / NC Manager' + the counter chips '3 total · 3 open' (using seeded data)
- WAIT** Wait for the loading spinner to disappear (data fetched from /api/deviations)
→ Three rows render: DEV-DEMO-001 OOS dissolution, DEV-DEMO-002 calibration drift, DEV-DEMO-003 viable contamination

